A Flexible Electromechanical Sensing Platform for In Vitro Hypoxic Cardiac Tissue Models

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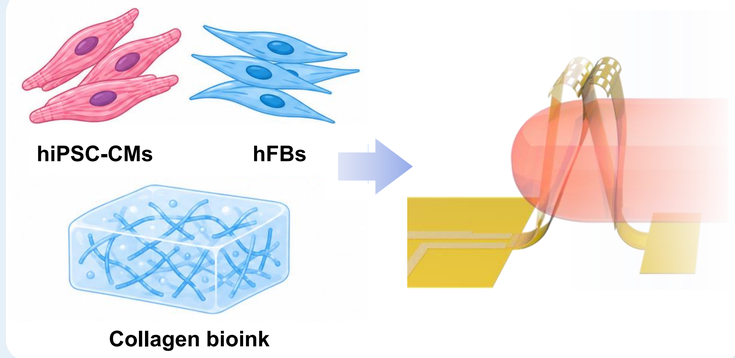
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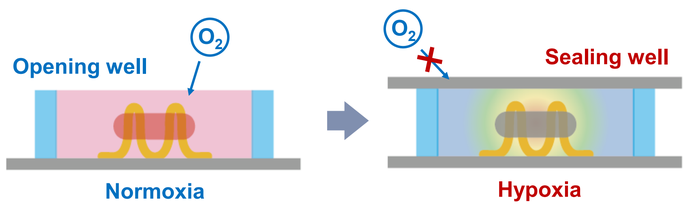
Poster: T061.e

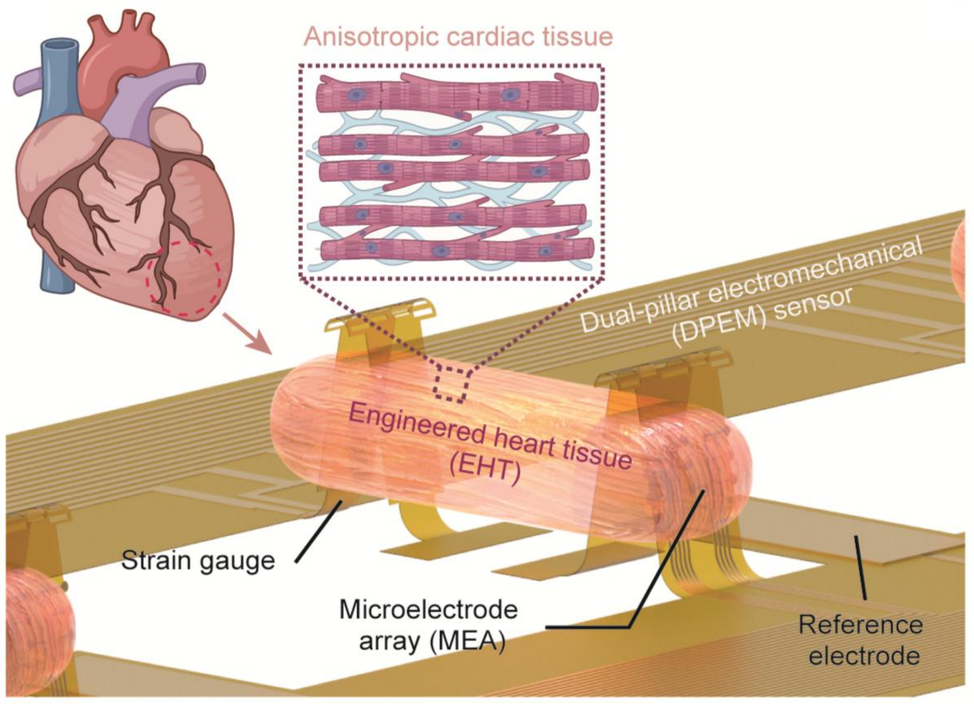
1 Graphical Abstract / Platform Overview

Engineered heart tissue (EHT) culture



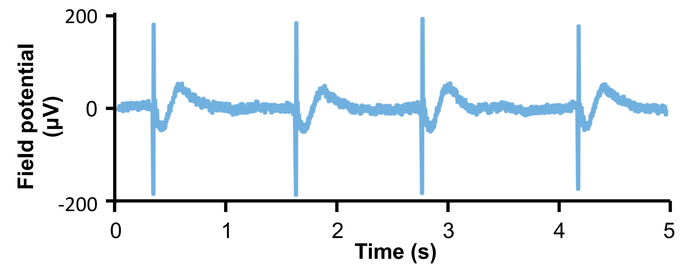
Hypoxia method



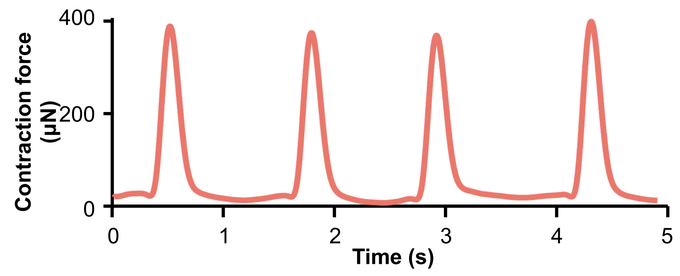


Simultaneous Elctromechanical measurement

Electrical activity (extracellular field potential)



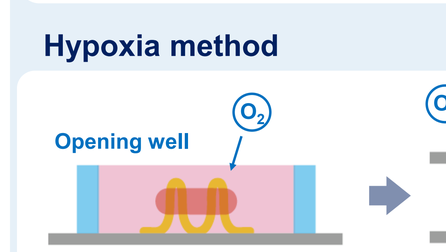
Mechanical contraction (strain / force)



2 Background & Motivation

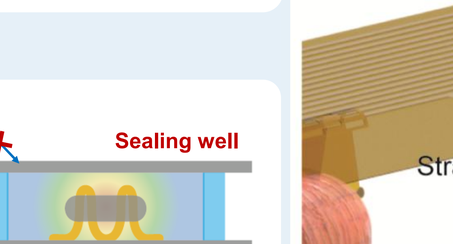
- Hypoxia is a major pathological driver in myocardial infarction, ischemic cardiomyopathy, and related cardiovascular diseases.
- Conventional *in vitro* platforms usually monitor only electrical activity or only mechanical contraction.
- Synchronized electromechanical monitoring is needed to evaluate excitation–contraction coupling under oxygen deprivation.

Health / Normoxia



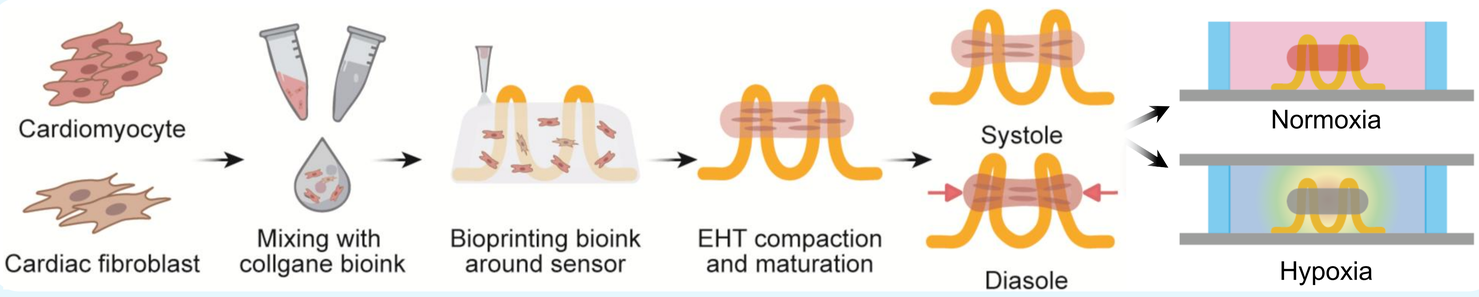
- Aligned cardiomyocytes
- Strong contraction
- Regular electrical activity
- Normal excitation–contraction coupling

Hypoxia / ischemia



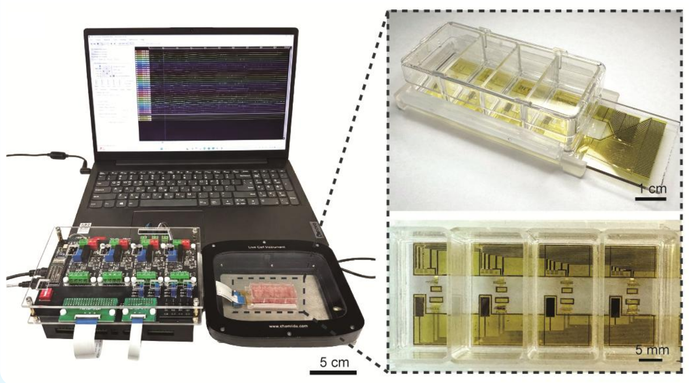
- Disorganized structure
- Reduced contraction
- Altered electrical activity
- Impaired excitation–contraction coupling

3 Engineered tissue formation / Workflow

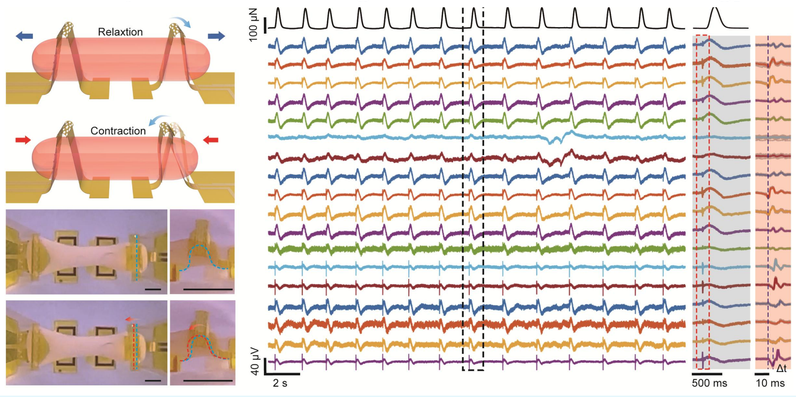


4 Simultaneous electromechanical recording

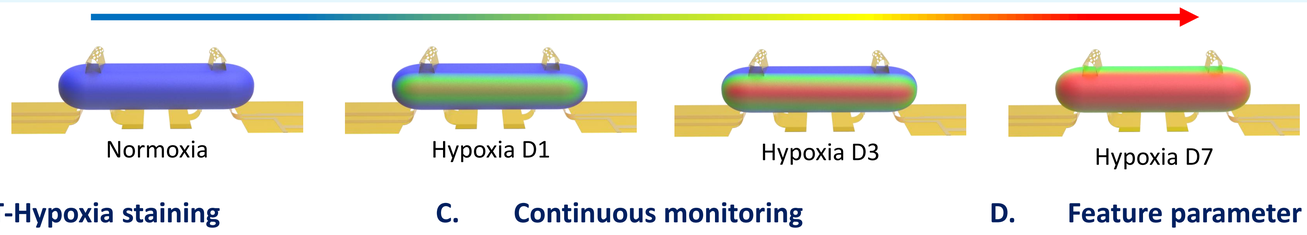
A. Measurement system and sensor array



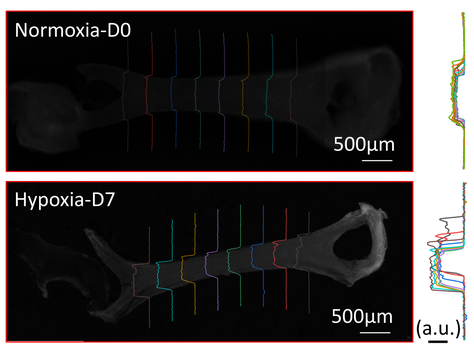
B. Synchronous beating and multi-channel recording]



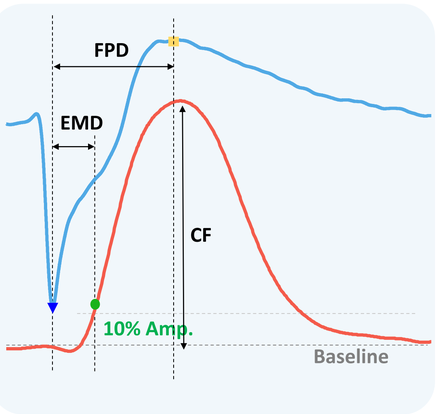
5 Hypoxia / EHT under hypoxic conditions.



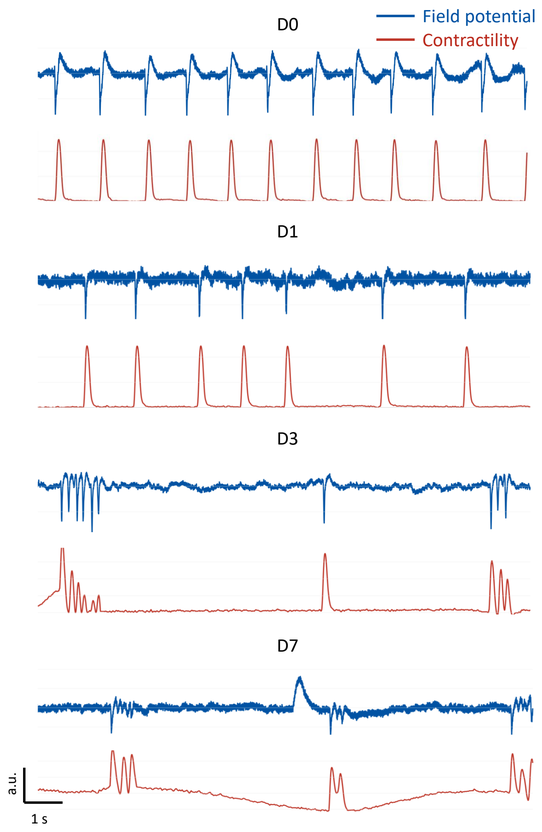
A. iT-Hypoxia staining



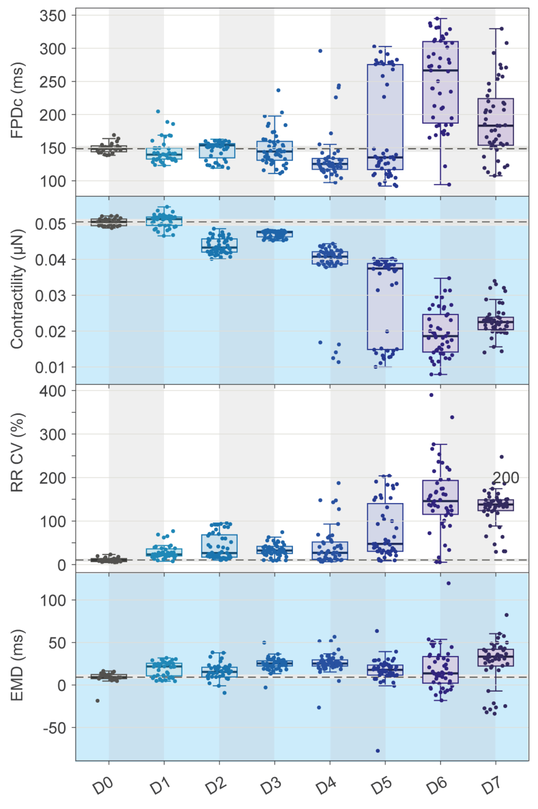
B. Feature analysis



C. Continuous monitoring



D. Feature parameter change



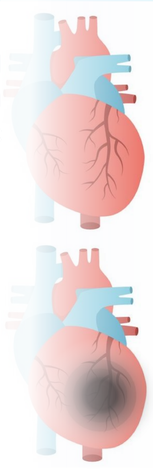
6 Key findings

- Integrated flexible electromechanical monitoring of engineered cardiac tissue.
- Reduced handling and minimized spatial mismatch between sensing modules (electrical and mechanical).
- Reproducible microenvironment for tissue formation, maturation, and longitudinal monitoring.
- Multimodal analysis provides richer information than single-parameter measurement.
- Useful for mechanistic studies, disease modeling, and screening of drugs or biomaterials targeting ischemic cardiac injury.

7 Conclusion & future outlook

This flexible electromechanical sensing platform provides an information-rich in vitro tool for hypoxia-related cardiac dysfunction studies.

- Mechanistic studies of ischemic cardiac injury
- Disease modeling using patient-specific cells
- Screening of drugs and biomaterials for cardiac repair



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